

Question 1: Object-Oriented Programming (OOP)**[10 marks]****Introduction: (2 marks)**

Object-Oriented Programming (OOP) is a programming paradigm that organizes software design around objects and data, rather than functions and logic. An object is a self-contained unit that bundles both data (attributes) and behavior (methods). OOP was developed to make large software systems easier to manage, maintain, and scale. Languages like Java, C++, Python, and C# are built on OOP principles.

Core Concepts: (4 marks)

OOP is built on four fundamental principles. Encapsulation bundles data and methods inside a class and restricts direct access to internal state using access modifiers (private, protected, public), exposing only what is necessary. Inheritance allows a child class to acquire properties and methods from a parent class, enabling code reuse and hierarchical relationships. For example, a Dog class can inherit from an Animal class. Polymorphism allows the same method name to behave differently in different contexts, achieved through method overloading (compile-time) and method overriding (runtime). Abstraction hides complex implementation details and exposes only the essential interface, implemented using abstract classes and interfaces.

Examples: (2 marks)

Consider a BankAccount class: it encapsulates balance and account number (private), exposes deposit() and withdraw() methods (public). A SavingsAccount can inherit from BankAccount and override calculateInterest() · this is both inheritance and polymorphism. In a vehicle system: Vehicle is the abstract base class; Car, Truck, Motorcycle inherit from it. Each overrides the move() method differently. A fleet management system can call move() on any vehicle without knowing its specific type · runtime polymorphism.

Conclusion: (2 marks)

OOP provides a structured approach to software development that mirrors real-world entities. Its principles of encapsulation, inheritance, polymorphism, and abstraction reduce code duplication, improve maintainability, and allow teams to build modular, scalable applications. OOP is the dominant paradigm in enterprise and system software development.

Question 2: Database Normalization**[10 marks]****Introduction: (2 marks)**

Database normalization is the process of organizing a relational database to reduce data redundancy and improve data integrity. It involves decomposing tables into smaller, well-structured tables and defining relationships between them. Normalization follows a series of rules called Normal Forms (NF). The goal is to eliminate anomalies that occur during insert, update, and delete operations on unnormalized data.

Core Concepts: (4 marks)

First Normal Form (1NF) requires that each table cell contains a single (atomic) value

and each record is unique · no repeating groups or arrays in columns. Second Normal Form (2NF) requires 1NF plus every non-key attribute must be fully functionally dependent on the entire primary key · eliminates partial dependencies, applicable when the key is composite. Third Normal Form (3NF) requires 2NF plus no transitive dependencies · a non-key attribute must not depend on another non-key attribute. Boyce-Codd Normal Form (BCNF) is a stricter version of 3NF where every determinant must be a candidate key. Fourth and Fifth Normal Forms address multi-valued and join dependencies respectively.

Examples: (2 marks)

Consider a Student_Course table: (StudentID, CourseID, StudentName, CourseName, InstructorName). This violates 2NF because StudentName depends only on StudentID (partial dependency) and CourseName depends only on CourseID. Decompose into: Student(StudentID, StudentName), Course(CourseID, CourseName, InstructorID), Enrollment(StudentID, CourseID), Instructor(InstructorID, InstructorName). This eliminates update anomaly (changing a student name requires one update), insert anomaly (can add a course without enrollments), and delete anomaly.

Conclusion: (2 marks)

Normalization is a systematic technique that ensures relational databases are free of redundancy and anomalies. While higher normal forms improve data integrity, they may require more joins in queries, sometimes causing denormalization for performance in read-heavy systems like data warehouses. Understanding normalization is fundamental to designing robust and efficient database schemas.

Question 3: Operating System Process Management

[10 marks]

Introduction: (2 marks)

Process management is one of the core functions of an operating system. A process is an instance of a program in execution, consisting of the program code, current activity (program counter, registers), process stack, and heap. The OS manages multiple processes simultaneously, allocating CPU time, memory, and I/O resources while ensuring isolation and synchronization between concurrent processes.

Core Concepts: (4 marks)

A process passes through five states during its lifecycle: New (being created), Ready (waiting for CPU), Running (executing on CPU), Waiting/Blocked (waiting for I/O or event), and Terminated (finished execution). The Process Control Block (PCB) stores all information about a process: PID, state, program counter, CPU registers, memory limits, open file list, and scheduling information. CPU scheduling algorithms determine which ready process gets the CPU: FCFS (First Come First Served) · non-preemptive, simple but causes convoy effect; SJF (Shortest Job First) · optimal average waiting time but requires future knowledge; Round Robin · preemptive, time-quantum based, fair for interactive systems; Priority Scheduling · each process has a priority, risk of starvation solved by aging.

Examples: (2 marks)

In Round Robin with quantum = 4ms: processes P1(24ms), P2(3ms), P3(3ms) execute in order P1(4), P2(3), P3(3), P1(4), P1(4), P1(4), P1(4), P1(4). Average waiting time = $(6+4+7)/3 = 5.67$ ms. Context switching overhead: when the OS switches from P1 to P2, it saves P1's

PCB (registers, program counter, stack pointer) and loads P2's PCB · this takes ~1-2 microseconds on modern hardware. Deadlock example: P1 holds Resource A and waits for B; P2 holds B and waits for A · circular wait. Prevented using Banker's Algorithm for safe state detection.

Conclusion: (2 marks)

Effective process management is essential for system performance, responsiveness, and stability. The OS must balance CPU utilization, throughput, turnaround time, waiting time, and response time. Modern OS kernels use multilevel feedback queue schedulers that adapt to process behavior, and multicore systems further require parallel scheduling across multiple CPUs with cache affinity consideration.

Question 4: Computer Networks and the OSI Model

[10 marks]

Introduction: (2 marks)

A computer network is a collection of interconnected devices that share resources and communicate using standardized protocols. Networks range from small Local Area Networks (LAN) to global Wide Area Networks (WAN) like the Internet. The OSI (Open Systems Interconnection) model, developed by ISO in 1984, provides a seven-layer conceptual framework that standardizes network communication functions, allowing interoperability between products from different vendors.

Core Concepts: (4 marks)

The OSI model has seven layers, each with distinct responsibilities: Layer 7 · Application: provides network services to end-user applications (HTTP, FTP, SMTP, DNS). Layer 6 · Presentation: data translation, encryption, compression (SSL/TLS, JPEG, ASCII). Layer 5 · Session: establishes, maintains, terminates sessions between applications. Layer 4 · Transport: end-to-end communication, segmentation, flow control, error recovery. TCP provides reliable, ordered delivery; UDP provides fast, connectionless delivery. Layer 3 · Network: logical addressing (IP), routing between networks using routers. Layer 2 · Data Link: framing, MAC addressing, error detection (Ethernet, Wi-Fi). Layer 1 · Physical: transmits raw bits over physical medium (cables, radio waves).

Examples: (2 marks)

When you access www.example.com: DNS (Application layer) resolves the domain to IP. Your browser creates an HTTP request (Application), TLS encrypts it (Presentation), TCP segments it with port 443 (Transport), IP adds source/destination addresses and routes through routers (Network), Ethernet frames carry it across the LAN with MAC addresses (Data Link), and electrical/optical signals physically transmit it (Physical). At the server, the process reverses · each layer strips its header and passes data up. TCP three-way handshake: SYN · SYN-ACK · ACK before data transfer.

Conclusion: (2 marks)

The OSI model provides a universal framework for understanding and designing network communication. While the practical Internet uses the simpler TCP/IP model (4 layers), OSI remains the standard reference for troubleshooting and protocol design. Understanding how data encapsulation works across layers is fundamental to networking, cybersecurity, and distributed systems engineering.

